

Topics: Inscribed Angles · Lines · Substitution · Isosceles Area ·
In-Class Test (25 Questions, 100 pts)

Instructions — Show clear work. Exact values (keep π , radicals) unless asked to round. Each question is worth **4 points**.

A. Inscribed Angles

1. In circle O , arc $\widehat{AB}$ measures 86° . Find $\angle ACB$ where C lies on the circle. (4)

2. In circle O , $\angle APB$ is an inscribed angle that intercepts arc $\widehat{AB}$ measuring 144° . Find $\angle APB$. (4)

3. In a circle, an inscribed angle measures 27° . What is the measure of its intercepted arc? (4)

4. In circle O , two inscribed angles intercept the same arc. One measures 41° . Find the other. (4)

5. In circle O , $\angle ACB$ is an inscribed angle and $\angle AOB$ is the central angle over the same arc. If $\angle AOB = 124^\circ$, find $\angle ACB$. (4)

B. Equation of a Line, Slope, Intercepts

6. Find the slope of the line through $(-6, 5)$ and $(2, -3)$. (4)
7. A line has slope $-\frac{3}{2}$ and passes through $(4, 1)$. Write its equation in slope-intercept form. (4)
8. Find the y -intercept of the line $3x - 4y = 20$. (4)
9. Determine the equation of the line with x -intercept -5 and y -intercept 4 (write in slope-intercept form). (4)
10. Lines $\ell_1 : y = 2x - 7$ and $\ell_2 : y = -\frac{1}{2}x + 1$. Are they parallel, perpendicular, or neither? Justify. (4)

C. Substitution (Solve by Substitution)

$$11. \begin{cases} y = 3x - 5 \\ 2x + y = 10 \end{cases} \quad (4)$$

$$12. \begin{cases} y = -x + 8 \\ 4x - 2y = 6 \end{cases} \quad (4)$$

$$13. \begin{cases} 5x - 2y = 11 \\ y = 2x + 1 \end{cases} \quad (4)$$

$$14. \begin{cases} y = \frac{1}{2}x - 3 \\ 3x + 2y = 4 \end{cases} \quad (4)$$

15. $\begin{cases} y = 4 - 3x \\ 6x + 2y = 1 \end{cases}$ (4)

D. Area of an Isosceles Triangle (Using the Median)

16. An isosceles triangle has equal sides 10 and base 12. Find the area using the median to the base. (4)

17. In $\triangle ABC$, $AB = AC = 13$ and $BC = 10$. Find the height from A to $\overline{BC}$ and the area. (4)

18. An isosceles triangle has base $b = 16$ and equal sides $s = 20$. Find its area. (4)

19. In an isosceles triangle, the area is 96 and the base is 12. Find the length of each equal (4)

side.

20. An isosceles triangle has perimeter 64 with equal sides x and base $b = 16$. Its area is 120. Find x . (4)

E. Elimination (Solve by Elimination)

21.
$$\begin{cases} 2x + 3y = 7 \\ 4x - 3y = 1 \end{cases} \quad (4)$$

22.
$$\begin{cases} 3x - 5y = 11 \\ 6x + 5y = 19 \end{cases} \quad (4)$$

$$23. \begin{cases} 7x + 2y = 5 \\ 5x - 2y = 19 \end{cases} \quad (4)$$

$$24. \begin{cases} 4x + 6y = 8 \\ -6x + 9y = 39 \end{cases} \quad (4)$$

$$25. \begin{cases} 5x + 4y = 1 \\ -10x + 6y = 44 \end{cases} \quad (4)$$